

The OK Is Not Enough

A Large-Scale Study of Consent Dialogs in Smartphone Applications
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1. Core Concept

The paper investigates whether privacy consent dialogs in Android and iOS applications actually give users a real choice before collecting personal data. The researchers set out to answer four main questions:

- How many mobile apps display privacy consent dialogs?
- Are these dialogs compliant with GDPR requirements?
- Do apps use dark patterns to push users into accepting?
- Do apps wait for user consent before sending personal data to trackers?

To answer these questions, they analyzed 3,006 Android apps and 1,773 iOS apps using an automated framework that detects consent dialogs, interacts with them, and monitors network traffic.

2. Method

The researchers built a complete automated analysis system, carried out in six steps.

Step 1: App Collection

- Downloaded 3,006 Android apps and 1,773 iOS apps.
- Selected top apps from many categories in both app stores.

Step 2: Static Analysis

They checked whether apps contained Consent Management Platforms (CMPs) such as IAB TCF. Result: only 5.1% of apps contained CMP libraries.

Step 3: Dynamic Analysis

For every app they:

- Installed it
- Opened it automatically
- Captured all network traffic
- Detected consent dialogs
- Clicked Accept or Reject
- Compared data sent before and after consent

Step 4: Consent Dialog Classification

Dialogs were classified into three categories:

- **Link** — only links to the privacy policy
- **Notice** — informs users but provides no real choice
- **Proper Dialog** — provides buttons like Accept/Reject

Step 5: Dark Pattern Detection

They checked whether dialogs:

- Had clear button labels
- Allowed easy rejection
- Highlighted the Accept button
- Required extra clicks to reject
- Forced users toward accepting

Step 6: Traffic Analysis

They monitored whether apps sent personal information:

- Before consent
- After accepting
- After rejecting

3. Key Findings

A. Very few apps show consent dialogs

Out of 3,654 successfully analyzed apps:

- 77.3% had no consent dialog
- 6.3% only showed a privacy link
- 4.5% showed a notice
- Only 11.9% showed a proper consent dialog

B. Nearly all consent dialogs used dark patterns

Among the proper dialogs, 98.8% violated at least one design requirement (Android: 98.4%, iOS: 99.5%). Common problems included:

- Accept button highlighted
- Reject button missing
- Rejecting required more clicks
- Ambiguous wording such as "Continue"

C. Apps send data before consent

- 35.2% of apps transmitted data containing unique identifiers before any user interaction
- 82.5% contacted at least one known tracker before the user interacted with a consent dialog

D. Consent often had little effect

Even after rejecting consent, some apps still transmitted pseudonymous data, and most tracking occurred before the user could choose.

E. CMP adoption was very low

- Only 5.1% of apps contained Consent Management Platform libraries

- Only 4.0% actively used IAB TCF settings

This means most developers use their own consent implementations.

F. Consent dialogs often failed GDPR expectations

The authors estimated that 43.1% of proper dialogs likely failed to obtain valid consent while still collecting personal data.

4. Full Summary

This paper examines whether smartphone applications genuinely respect users' privacy choices. Although the GDPR requires user consent before tracking or collecting personal data, the authors found that most mobile apps do not follow these principles.

The researchers created an automated framework capable of downloading apps, launching them, detecting privacy consent dialogs, interacting with those dialogs, and monitoring all network traffic. They applied this framework to 3,006 Android apps and 1,773 iOS apps, making it one of the largest studies of mobile consent dialogs.

Their analysis revealed that only about 22.3% of apps displayed any form of privacy consent dialog, and only 11.9% offered users an actual dialog with actionable choices. Even among these, almost every dialog used at least one dark pattern, such as emphasizing the Accept button, hiding or complicating the Reject option, or using unclear wording that nudged users toward accepting.

The network analysis showed an even more concerning result: many apps transmitted personal or pseudonymous information to third-party trackers before users had any opportunity to provide consent. This indicates that the consent dialogs often served as a formality rather than a genuine privacy control.

The study also found that Consent Management Platforms (CMPs) were rarely used in mobile apps, meaning most developers implemented their own consent mechanisms, which frequently failed to meet GDPR expectations.

Overall, the authors conclude that mobile privacy consent dialogs are largely ineffective. They argue that many apps ignore meaningful consent, rely on manipulative interface designs, and collect personal data before users can make an informed decision. Their findings suggest that stronger enforcement, better dialog design, and stricter compliance measures are necessary to protect user privacy in mobile applications.